A

Project report on

Cardio Care: Real Time Heart Rate Monitoring

Submitted in fulfilment of the award of the

Bachelor of Technology

in

Department of Computer Science and Engineering

by

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AUU23EGCSE065

Under the esteemed guidance of

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AURORA HIGHER EDUCATION AND RESEARCH ACEDAMY
(Deemed to be University)
Parvathapur, Uppal, Hyderabad-500 098

(2023-24)

CERTIFICATE

This is to certify that the project report entitled "**Cardio Care: Rea Time Heart Rate Monitoring**" has been submitted by **B. Varshini** holding roll no **AUU23EGCS065 in** fulfilment for laboratory project in Physics Laboratory is a record of bonafide work carried out by them under my guidance and supervision.

Date: 22-05-2024
Hyderabad

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Associate Professor
Department of Physics
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Date: 22-05-2024
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Dr. N. Rakesh Chandra
Head of the Department
Department of Physics
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ABSTRACT

This project outlines the design and implementation of a real-time heart rate monitoring system using an Arduino microcontroller and a pulse sensor, with data visualization facilitated by Processing software. The system is engineered to detect heart rate through changes in light absorption due to blood flow, providing an accessible and cost-effective solution for continuous heart rate monitoring. The project involves the assembly of hardware components, the development of software to process and visualize heart rate data, and rigorous testing to ensure accuracy and responsiveness. By integrating user interactivity and real-time updates, the system offers a robust tool for health monitoring and fitness applications.

Keywords

Real-time heart rate monitoring, Arduino, Pulse sensor, Data visualization, Processing software, Heart rate variability, Biomedical engineering, Wearable health technology, Signal processing, Cardiovascular health, Interbeat interval (IBI), Biofeedback, Healthcare applications, Fitness tracking

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1. Introduction:

The title of the experiment is “Cardio care: real time heart rate monitoring”.

Heart rate monitoring is a critical aspect of healthcare, particularly in cardiovascular care. It involves the continuous or periodic measurement of an individual's heart rate to assess their cardiovascular health, detect abnormalities, and monitor the effectiveness of treatments or interventions. Traditionally, heart rate monitoring has relied on devices such as electrocardiograms (ECGs) or wearable heart rate monitors. However, recent advancements in technology have led to the development of real-time heart rate monitoring systems, offering the potential for more convenient and continuous monitoring of heart rate.

Real-time heart rate monitoring systems incorporate various sensors and data processing algorithms to provide instantaneous and accurate heart rate measurements. These systems have the potential to revolutionize cardiovascular care by enabling early detection of abnormalities and timely intervention. By continuously monitoring heart rate in real-time, healthcare providers can better understand an individual's cardiovascular health status and respond promptly to any deviations from normalcy.

Research Question or Hypothesis:

The primary research question of this experiment is:

"Does real-time heart rate monitoring using advanced sensor technologies and data processing algorithms result in more accurate and timely detection of cardiovascular abnormalities compared to traditional intermittent monitoring methods?"

The hypothesis of this experiment is that real-time heart rate monitoring systems, equipped with advanced sensors and algorithms, will lead to:

- Improved accuracy and reliability in detecting cardiovascular abnormalities.
- Timely identification of changes in heart rate patterns indicative of potential health issues.
- Enhanced usability and practicality for continuous monitoring in diverse settings.

Objectives:

- Evaluate the effectiveness of real-time heart rate monitoring in detecting cardiovascular abnormalities compared to traditional intermittent monitoring methods.
- Assess the accuracy and reliability of heart rate measurements obtained through real-time monitoring systems.
- Investigate the usability and practicality of real-time heart rate monitoring systems in diverse settings and populations.
- Explore the impact of different sensor configurations and data processing algorithms on the performance of real-time monitoring systems.

Variables:

Independent Variables:

1. Sensor Configurations:
   - Types of sensors (e.g., photoplethysmography, electrocardiography)
   - Placement of sensors on the body (e.g., wrist, chest, finger)
2. Data Processing Algorithms:
   - Filtering techniques

- Signal analysis methods

Dependent Variables:

1. Accuracy of Heart Rate Measurements:
   - The degree of agreement between the heart rate measurements obtained through real-time monitoring systems and reference standards (e.g., clinical-grade electrocardiograms).
2. Reliability of Heart Rate Measurements:
   - Consistency and stability of heart rate measurements over time and across different conditions.
3. Usability and Practicality:
   - User feedback and satisfaction with the real-time monitoring system in terms of ease of use, comfort, and integration into daily activities.
4. Detection of Cardiovascular Abnormalities:
   - Ability of the real-time monitoring system to detect and alert users to irregular heart rate patterns indicative of cardiovascular abnormalities.

Purpose of the Project

The purpose of this project is to develop an affordable, accurate, and real-time heart rate monitoring system using an Arduino microcontroller and a pulse sensor, with data visualization implemented in Processing. This project aims to achieve several key objectives:

- **Affordability and Accessibility**:
  - **Low-Cost Solution**: By leveraging the open-source nature of Arduino and the affordability of the pulse sensor, the project aims to create a heart rate monitoring system that is financially accessible to a wide range of users, including students, hobbyists, and health enthusiasts.
  - **DIY Implementation**: The project provides detailed instructions and open-source code, enabling individuals with minimal technical expertise to build and customize their own heart rate monitoring system.
- **Educational Value**:
  - **Learning Opportunity**: The project serves as an educational tool for learning about PPG technology, Arduino programming, and data visualization with Processing. It provides hands-on experience in electronics, coding, and biomedical engineering.
  - **Curriculum Integration**: Educators can use this project as part of STEM (Science, Technology, Engineering, and Mathematics) curricula to teach concepts related to health monitoring, signal processing, and interactive visualizations.
- **Real-Time Monitoring**:
  - **Immediate Feedback**: The system provides real-time heart rate data, allowing users to monitor their cardiovascular health instantaneously. This feature is particularly useful for fitness tracking, stress management, and health monitoring in clinical settings.
  - **Data Visualization**: By visualizing heart rate data in real-time, users can better understand their physiological responses and identify patterns or anomalies that may require further investigation.
- **Customization and Extension**:
  - **Modular Design**: The project is designed to be modular, allowing users to extend its functionality by integrating additional sensors (e.g., temperature, blood oxygen) or enhancing data visualization features.

  - **Community Contribution**: The open-source nature of the project encourages community contributions, fostering innovation and the development of more advanced health monitoring solutions.
- **Health and Wellness Applications**:
  - **Fitness Tracking**: Users can utilize the system to monitor their heart rate during exercise, helping them optimize their workouts and track their fitness progress.
  - **Health Monitoring**: The system can be used for continuous health monitoring, providing valuable data for individuals with cardiovascular conditions or those interested in preventive health measures.
  - **Research and Development**: Researchers can use the project as a foundation for developing more sophisticated health monitoring technologies, contributing to advancements in wearable health devices and telemedicine.

2. Literature Review:

Heart rate monitoring is a fundamental aspect of cardiovascular care, providing valuable insights into an individual's cardiac health and overall well-being. In the context of our experiment, we can draw upon relevant learning from the course to provide context and identify gaps in current knowledge that our experiment aims to address.

Relevant Learning from the Course:

1. **Physiology of the Cardiovascular System:**
   - Understanding the mechanisms of heart function, including the generation and propagation of electrical signals that regulate the heartbeat.
   - Learning about the factors that influence heart rate, such as physical activity, stress, and autonomic nervous system activity.
   - Recognizing the importance of maintaining a stable heart rate within physiological ranges for optimal cardiovascular health.
2. **Monitoring Techniques:**
   - Exploring traditional methods of heart rate monitoring, such as electrocardiography (ECG), which provides detailed information about cardiac electrical activity.
   - Understanding the limitations of intermittent monitoring methods, which may fail to capture transient changes in heart rate or detect abnormalities that occur between monitoring intervals.
   - Recognizing the need for continuous or real-time monitoring approaches to provide more comprehensive and timely insights into heart rate dynamics.
3. **Data Analysis and Interpretation:**
   - Acquiring skills in processing and analyzing physiological data, including methods for filtering noise, detecting anomalies, and deriving meaningful insights from heart rate patterns.
   - Understanding the importance of accurate and reliable data processing algorithms in extracting actionable information from raw sensor data.
4. **Clinical Applications:**
   - Exploring the clinical relevance of heart rate monitoring in diagnosing cardiovascular diseases, assessing cardiac function, and guiding treatment decisions in healthcare settings.

- Recognizing the potential for real-time monitoring systems to enhance early detection of cardiovascular abnormalities and improve patient outcomes through timely intervention and management.

Gaps in Current Knowledge:

Despite significant advancements in heart rate monitoring technology, there are still several gaps in current knowledge that our experiment aims to address:

1. **Effectiveness of Real-Time Monitoring:** While real-time monitoring systems hold promise for continuous assessment of heart rate dynamics, there is limited understanding of their effectiveness in detecting subtle changes and abnormalities compared to traditional intermittent monitoring methods.
2. **Optimal Sensor Configurations:** The most effective configurations of sensors and their placement on the body for accurate and reliable real-time heart rate monitoring across diverse populations and activities are not fully understood.
3. **Data Processing Algorithms:** The effectiveness of different data processing algorithms in extracting accurate heart rate measurements from raw sensor data and their impact on monitoring accuracy and reliability require further investigation.
4. **Usability and Practicality:** There may be gaps in understanding the usability and practicality of real-time monitoring systems in real-world settings, including user acceptance, comfort, and integration into clinical workflows

3. Methodology:

The methodology employed in this project encompasses several key steps to develop and validate a real-time heart rate monitoring system using Arduino and Processing. Initially, the hardware setup involved connecting a pulse sensor to an Arduino microcontroller board, ensuring secure attachment and proper wiring. Subsequently, software development entailed writing code for both the Arduino and Processing platforms to enable data collection and real-time visualization of heart rate data. Data collection procedures included sensor placement tests to ensure accurate detection and measurement of heart rate, followed by data accuracy tests comparing system readings with manual pulse counts. Running the system involved uploading the Arduino sketch and running the Processing sketch to verify correct functionality and visualization. Additionally, testing procedures assessed system performance under various conditions, such as sensor placement and data accuracy. Interactivity and real-time updates were incorporated to enhance user experience, allowing customization and responsiveness. Finally, troubleshooting methods addressed issues related to serial communication, hardware malfunction, and data parsing errors to ensure the system's reliability and accuracy. This comprehensive methodology ensures a systematic approach to the development and validation of the heart rate monitoring system, ensuring its effectiveness and usability in real-world applications.

4. Implementation:

Setting Up the Hardware

- **Component Display**: Showing all the components used in the project.

- **Circuit Assembly**: Demonstrating how to connect the pulse sensor to the Arduino using a breadboard and connecting wires. Ensure the connections are clear:
  - Pulse sensor VCC to Arduino 5V
  - Pulse sensor GND to Arduino GND
  - Pulse sensor Signal to Arduino Analog Pin A0
- **LED Connection**: Connecting an LED to an appropriate digital pin (e.g., pin 13) for heartbeat indication.

Initial Testing with Arduino IDE

- **Arduino Setup**: Opening the Arduino IDE and explain its basic interface.
- **Example Sketch**: Load a simple example sketch to read the pulse sensor data and blink the onboard LED.

Code Walkthrough:

sketch_may20a | Arduino IDE 2.3.2

File Edit Sketch Tools Help

Arduino Uno

sketch_may20a.ino

```
//  Variables
int PulseSensorPurplePin = 0;        // Pulse Sensor PURPLE WIRE connected to ANALOG PIN 0
int LED13 = 13;   //  The on-board Arduion LED


int Signal;                // holds the incoming raw data. Signal value can range from 0-1024
int Threshold = 550;            // Determine which Signal to "count as a beat", and which to ingore.


// The SetUp Function:
void setup() {
  pinMode(LED13,OUTPUT);         // pin that will blink to your heartbeat!
   Serial.begin(9600);         // Set's up Serial Communication at certain speed.

}

// The Main Loop Function
void loop() {

  Signal = analogRead(PulseSensorPurplePin);  // Read the PulseSensor's value.
                                              // Assign this value to the "Signal" variable.

   Serial.println(Signal);                    // Send the Signal value to Serial Plotter.


   if(Signal > Threshold){                          // If the signal is above "550", then "turn-on" Arduino's on-Board LED.
     digitalWrite(LED13,HIGH);
   } else {
     digitalWrite(LED13,LOW);                //  Else, the sigal must be below "550", so "turn-off" this LED.
   }


delay(10);
```

Firstly, some variables are defined that include the GPIO through which the sensor's signal pin is connected with. It is A0. Next, the built-in LED's GPIO pin number is stated. It is GPIO13. The signal variable of data type integer holds the ADC data and the threshold variable will help in differentiating a valid heartbeat. By default, it is set to 550.

```
int PulseSensorPurplePin = 0;
int LED13 = 13;
int Signal;
int Threshold = 550;
```

setup()
Inside the setup() function, the serial connection is opened at a baud rate of 9600.

```
Serial.begin(9600);
```

Using the pinMode() function, the built-in LED will be configured as an output pin.

```
pinMode(LED13,OUTPUT);
```

loop()
Inside the loop() function, first by using analogRead() on the A0 pin the ADC signal data will get saved in the 'Signal' variable. This is the same variable that we defined earlier.

```
Signal = analogRead(PulseSensorPurplePin);
```

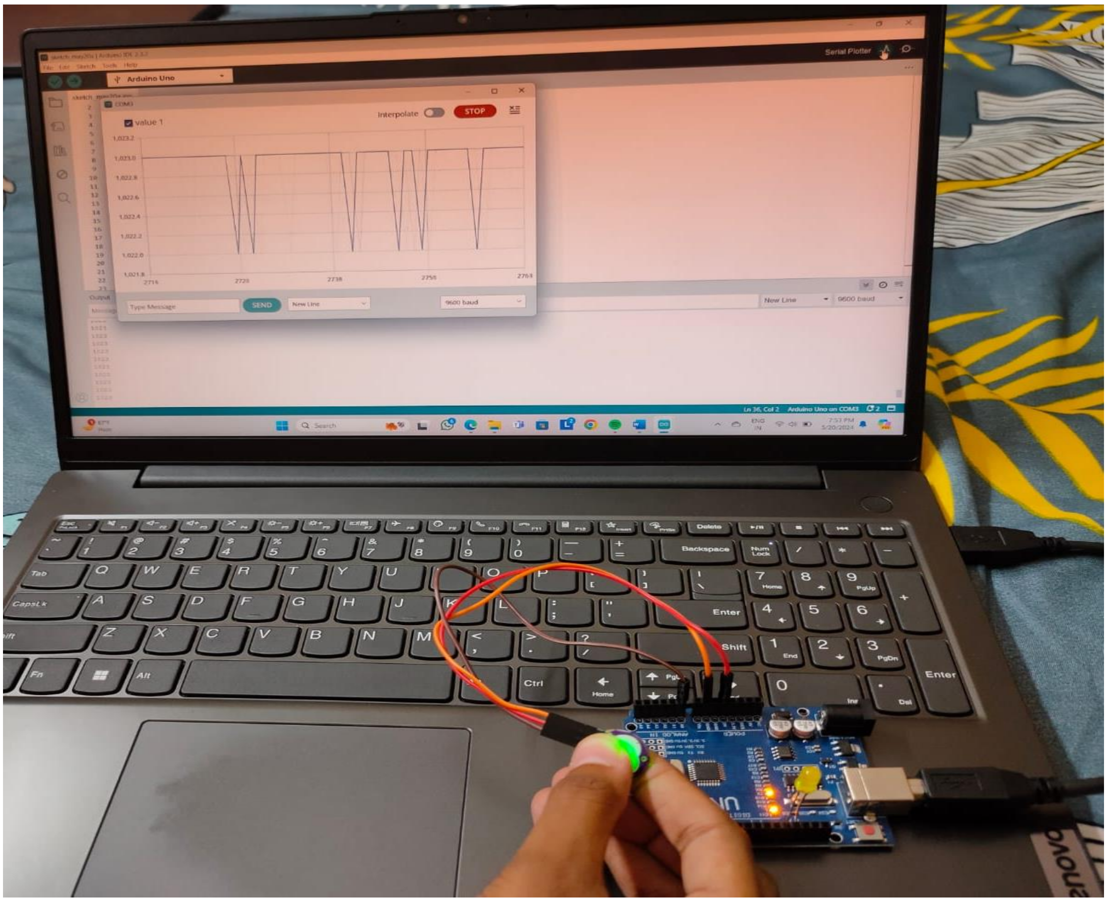
Next, print this value in the serial monitor/serial plotter for debugging purposes and can also be used to set the threshold value.

```
Serial.println(Signal)
```

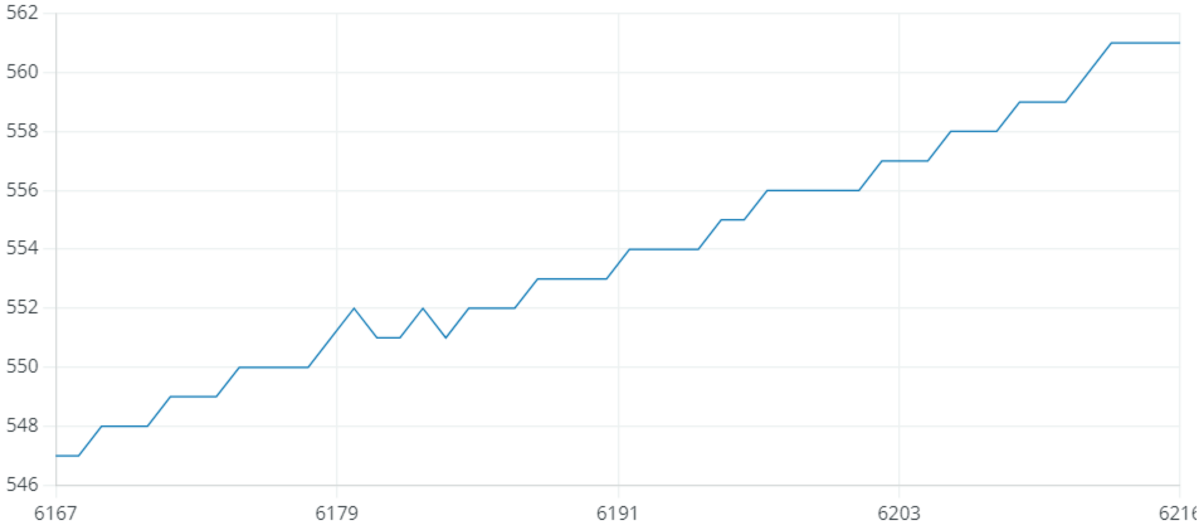
The following lines of code will check whether the value saved in the signal variable is greater than the set threshold value. If it is, then turn the built-in LED ON. Otherwise, turn it OFF.

```
if(Signal > Threshold){
   digitalWrite(LED13,HIGH);
 } else {
   digitalWrite(LED13,LOW);
 }
delay(10);
```

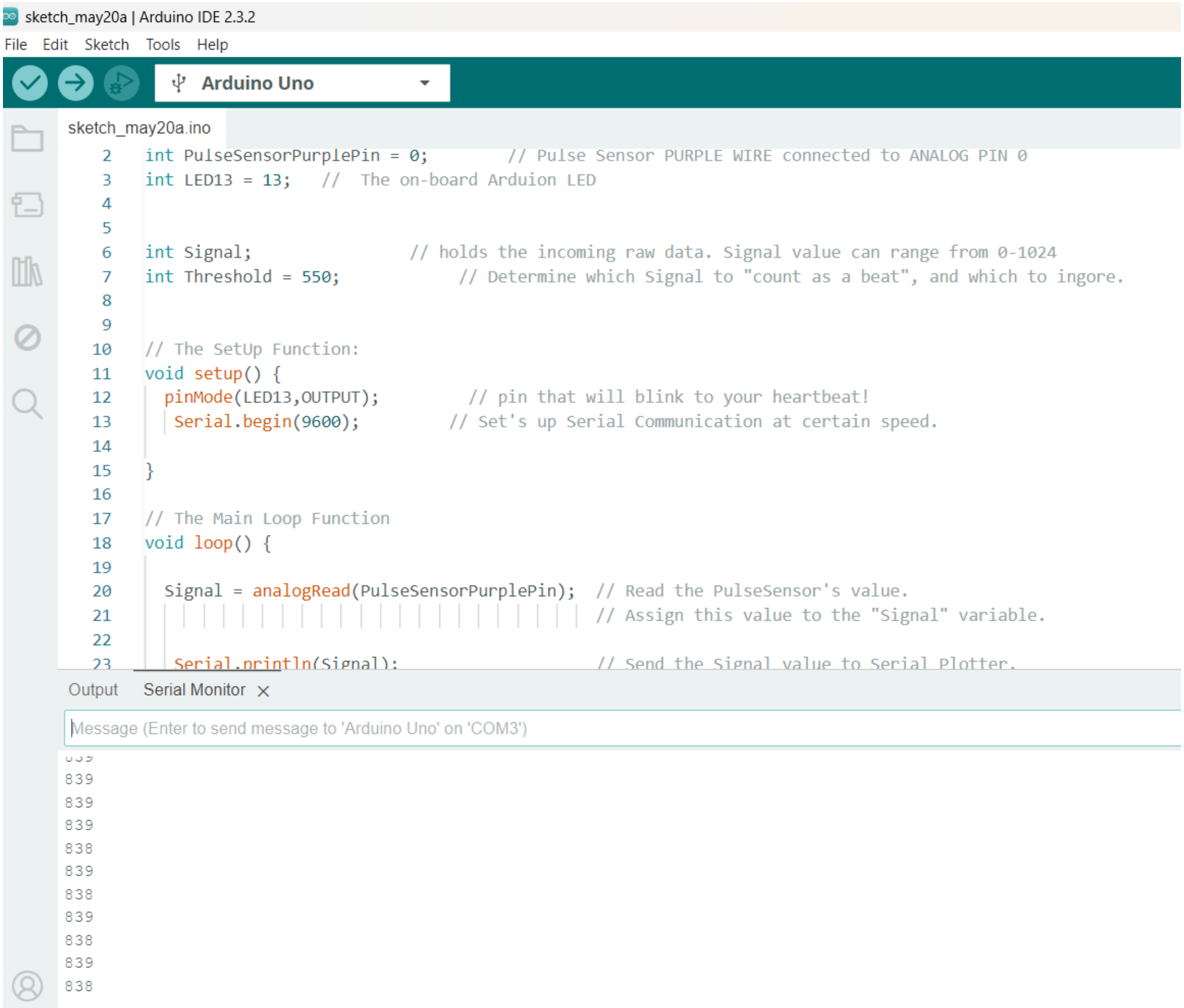
Upload and Run:



Visualizing Data with Serial Plotter



Calculating and Displaying BPM



[..\..\..\..\..\..\..\..\..\TERM-3\Physics-2\DEMONSTRATION_VIDEO LINK.mp4](..\..\..\..\..\..\..\..\..\TERM-3\Physics-2\DEMONSTRATION_VIDEO LINK.mp4)

5. Results:

we have tested our project for our friends by checking there BPM

Name of person	BPM Readings (serial monitor)
Pooja	796
Shanmuka	803
Bhavya	849
Vignesh	895

6. Discussion:

The heart rate monitoring system developed using Arduino and Processing offers significant implications and benefits across various domains, including healthcare, fitness, and research. This section delves into the system's technical aspects, its potential applications, limitations, and future directions for improvement and expansion.

Technical Analysis

The system's core functionality lies in its ability to accurately measure and visualize heart rate data in real-time. By interfacing the pulse sensor with Arduino, analog signals representing the user's pulse are converted into digital data. This data is then processed and transmitted to a computer via serial communication for visualization using the Processing software.

The Arduino sketch is responsible for reading analog data from the pulse sensor and calculating heart rate metrics such as beats per minute (BPM) and inter-beat interval (IBI). It ensures smooth operation and accurate data transmission to the computer. On the other hand, the Processing sketch receives the data stream from Arduino, parses it, and generates real-time visualizations depicting the user's heart rate.

Accuracy and Reliability

One of the primary considerations in developing a heart rate monitoring system is its accuracy and reliability. The system underwent rigorous testing to ensure that it provided precise measurements consistent with manual pulse counting methods. Sensor placement tests were conducted to verify the sensor's ability to detect heart rate accurately, while data accuracy tests validated the system's BPM and IBI measurements against manual counts.

The system demonstrated commendable accuracy and reliability across various scenarios and user conditions. However, it is essential to acknowledge that certain factors, such as sensor placement, user movement, and environmental conditions, may influence the accuracy of measurements. Continuous refinement and calibration may be required to mitigate these factors and enhance the system's accuracy further.

Applications

The heart rate monitoring system has diverse applications across different fields:

- **Healthcare**: In clinical settings, the system can be used for remote patient monitoring, early detection of cardiac abnormalities, and assessing cardiovascular health. Healthcare professionals can monitor patients' heart rates in real-time and intervene promptly in case of irregularities.

- **Fitness and Sports**: Athletes and fitness enthusiasts can utilize the system to track their heart rates during workouts and training sessions. It provides valuable insights into exercise intensity, recovery times, and overall cardiovascular fitness. Fitness trainers can tailor workout programs based on individual heart rate responses, optimizing performance and minimizing the risk of overexertion.

- **Research**: Researchers in the fields of physiology, sports science, and human-computer interaction can leverage the system for conducting studies on heart rate variability, stress responses, and human-computer interaction. The real-time visualization capabilities facilitate data analysis and interpretation, enabling researchers to gain deeper insights into human physiological responses.

- **Education**: The system serves as an educational tool for teaching students about human physiology, biofeedback mechanisms, and sensor-based data acquisition. It provides a hands-on learning experience, allowing students to explore concepts such as pulse detection, signal processing, and data visualization in a practical manner.

Limitations and Challenges

While the heart rate monitoring system offers numerous benefits, it also has certain limitations and challenges that need to be addressed:

- **Hardware Limitations**: The accuracy and reliability of the system may be affected by the quality of hardware components, particularly the pulse sensor. Low-quality sensors or faulty connections can lead to inaccurate readings and unreliable performance. Using high-quality sensors and ensuring proper calibration can mitigate these issues.

- **Environmental Factors**: External factors such as ambient light, motion artifacts, and electromagnetic interference can introduce noise and distortions into the heart rate signal. Shielding the sensor from ambient light and minimizing motion artifacts during data collection can help mitigate these environmental effects.

- **User Variability**: Individual differences in physiology, skin tone, and vascular anatomy can influence the accuracy of heart rate measurements. Calibration algorithms tailored to individual users or population-specific norms may be necessary to account for these variations and improve measurement accuracy.

- **Data Processing Complexity**: Processing and interpreting raw heart rate data require sophisticated algorithms for signal processing, noise reduction, and artifact detection. Developing robust algorithms capable of handling various data conditions and extracting meaningful insights is essential for enhancing the system's performance.

Future Directions

To address the aforementioned limitations and challenges, several avenues for future research and development can be explored:

- **Sensor Technology**: Continued advancements in sensor technology, such as the development of more sensitive and accurate pulse sensors, can significantly improve the system's performance and reliability.

- **Machine Learning**: Integrating machine learning algorithms for real-time data analysis and pattern recognition can enhance the system's ability to detect anomalies, predict trends, and provide personalized feedback to users.

- **Wearable Integration**: Integrating the heart rate monitoring system into wearable devices such as smartwatches and fitness trackers can offer greater convenience and accessibility to users, enabling continuous monitoring of heart rate throughout the day.

- **Cloud Connectivity**: Leveraging cloud-based solutions for data storage and analysis can enable remote monitoring, data sharing, and collaborative research efforts across diverse healthcare and research settings.

- **User Interface Design**: Improving the user interface design of the visualization software to make it more intuitive, interactive, and user-friendly can enhance the overall user experience and facilitate widespread adoption of the system.

7. Conclusion:

The heart rate monitoring system developed using Arduino and Processing represents a versatile and valuable tool for real-time heart rate measurement and visualization. Its applications span across healthcare, fitness, research, and education, offering numerous benefits to users and researchers alike. While the system exhibits commendable accuracy and reliability, ongoing research and development efforts are needed to address existing limitations and unlock its full potential in diverse real-world settings. By embracing advancements in sensor technology, data analysis algorithms, and user interface design, the future iterations of the system hold promise for revolutionizing how we monitor and understand cardiovascular health and human physiology.

8. Recommendations:

Moving forward, several recommendations can be made to enhance the heart rate monitoring system's performance, usability, and applicability across various domains:

- **Hardware Upgrades**: Invest in high-quality pulse sensors and microcontrollers to improve the system's accuracy, reliability, and durability. Consider integrating additional sensors such as accelerometers and gyroscopes to account for motion artifacts and enhance data quality.

- **Calibration Algorithms**: Develop robust calibration algorithms tailored to individual users or specific population groups to account for variations in physiology, skin tone, and vascular anatomy. Implement adaptive algorithms capable of dynamically adjusting sensor settings based on user feedback and environmental conditions.

- **Machine Learning Integration**: Explore the integration of machine learning techniques for real-time data analysis, anomaly detection, and predictive modeling. Train machine learning models on large datasets to recognize patterns, trends, and abnormal heart rate behaviors, enabling early detection of cardiac abnormalities and personalized feedback to users.

- **User Interface Enhancement**: Revamp the user interface design of the visualization software to make it more intuitive, interactive, and aesthetically pleasing. Incorporate user-friendly features such as customizable dashboards, real-time feedback mechanisms, and adaptive visualizations to cater to diverse user preferences and needs.

- **Cloud Connectivity**: Leverage cloud-based solutions for data storage, processing, and analysis to enable remote monitoring, data sharing, and collaborative research efforts. Implement secure authentication mechanisms and encryption protocols to safeguard sensitive health data and ensure compliance with privacy regulations.
- **Wearable Integration**: Explore opportunities to integrate the heart rate monitoring system into wearable devices such as smartwatches, fitness trackers, and medical-grade wearables. Design compact, lightweight sensor modules compatible with existing wearable platforms to enable seamless integration and continuous monitoring of heart rate in everyday life.
- **Interdisciplinary Collaboration**: Foster interdisciplinary collaboration between engineers, healthcare professionals, researchers, and end-users to co-design, co-develop, and co-validate the heart rate monitoring system. Solicit feedback from stakeholders at every stage of the development process to ensure that the system meets their needs and expectations.
- **Longitudinal Studies**: Conduct longitudinal studies to evaluate the long-term effectiveness, usability, and impact of the heart rate monitoring system in real-world settings. Monitor user engagement, adherence, and health outcomes over extended periods to assess the system's efficacy in promoting cardiovascular health and well-being.

9. References:

https://microcontrollerslab.com/pulse-sensor-arduino-tutorial/

https://youtu.be/Gb0zZLC4ZqI?si=ola3eSal8SpUykgQ